

1 Collection and Evaluation of Serial Numbers

General

Serial numbers are assigned to differentiate between individual material items.

Purpose

The MES provides the following options with corresponding specifications to collect and/or use serial numbers:

- Option 1: Enter serial numbers for operations that are not subject to batch management.
- Option 2: Enter serial numbers for operations that are subject to batch management.
 - Enter the serial number manually. This manually entered serial number is assigned to a HYDRA batch number.
 - The serial number is assigned automatically as HYDRA batch number. The system creates a batch with the assigned serial number.
 - The serial number is assigned automatically using a number range. The assigned serial number is assigned to a new HYDRA batch number.

The paragraphs that follow describe the logic and processing of serial number collection (depending on the configuration).

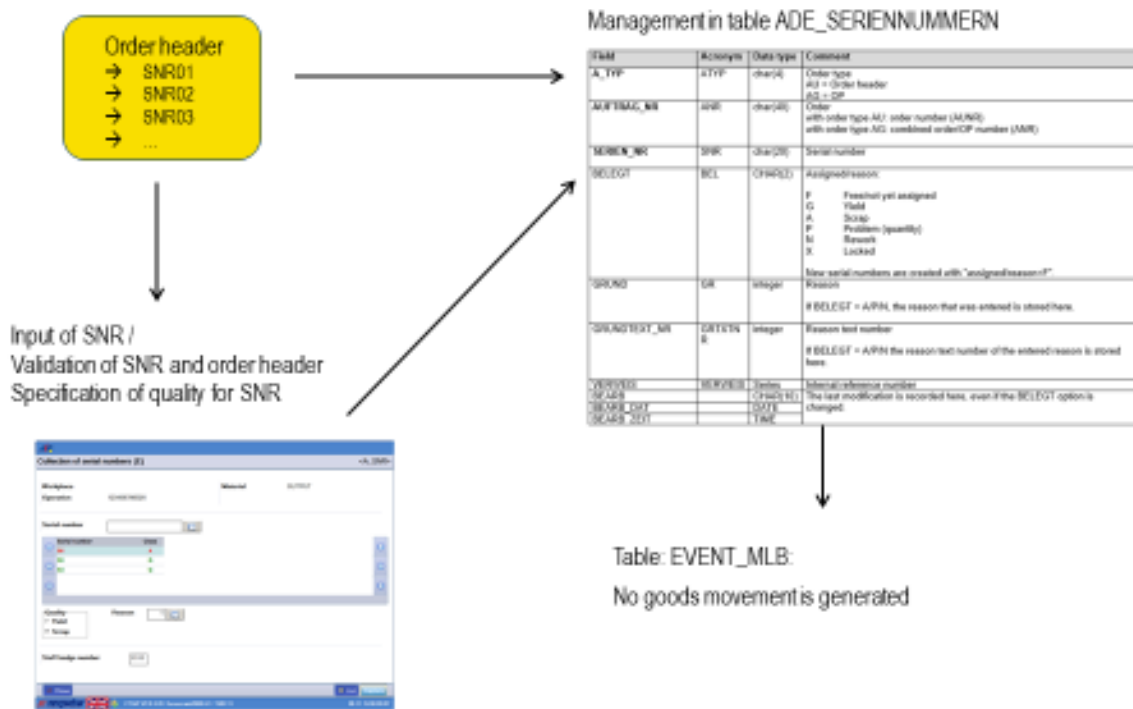
How to collect serial numbers for OPs that are not subject to batch management

The posted serial number is only recorded in the BDE module if the operation is not subject to batch management.

The serial number cannot be generated automatically. The user enters the serial number (manually or by scanning).

Goods movements are not generated per serial number.

OP not requiring batch management – A_SNR "E – manual input"



How to collect serial numbers for OPs that are subject to batch management

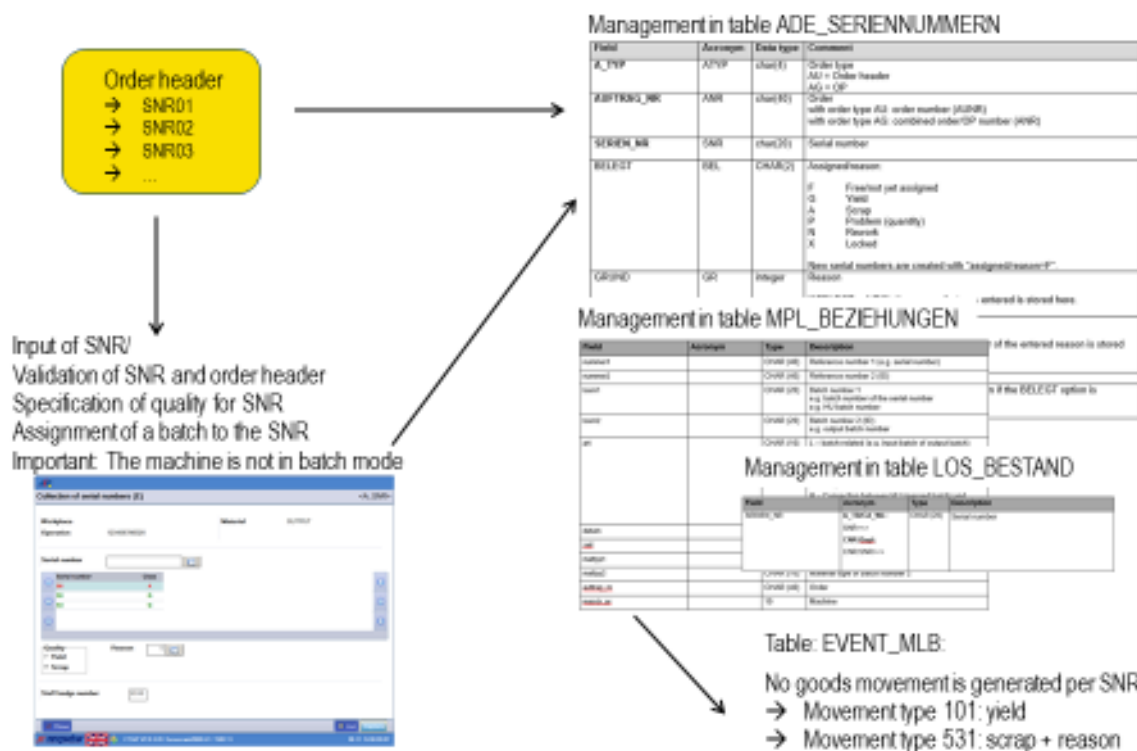
Collection of serial numbers (A SNR) including manual assignment of serial numbers - type "E"

An additional batch is created in the material and production logistics module for each registered serial number if the operation is subject to batch management.

The connection between the current output batch and the registered serial number is saved additionally in the database for traceability purposes. In this case, the output batch is considered as ID without inventory and, therefore, does not receive a quantity.

A goods movement is generated for each serial number (please note: the machine mustn't be in batch mode!).

OP requiring batch management – A_SNR "E – manual input"



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Collection of serial numbers (A_SNR) including automatic assignment of serial numbers - type "G"

If the option "serial number requirement = G" is set for the operation, output batches will be recorded as serial numbers. In this case, the serial number is the output batch number.

The "serial number" field is disabled and includes the current output batch. Once the badge number has been entered, click "capture" to post the data. Enter a valid scrap reason if the entered quality is not "yield".

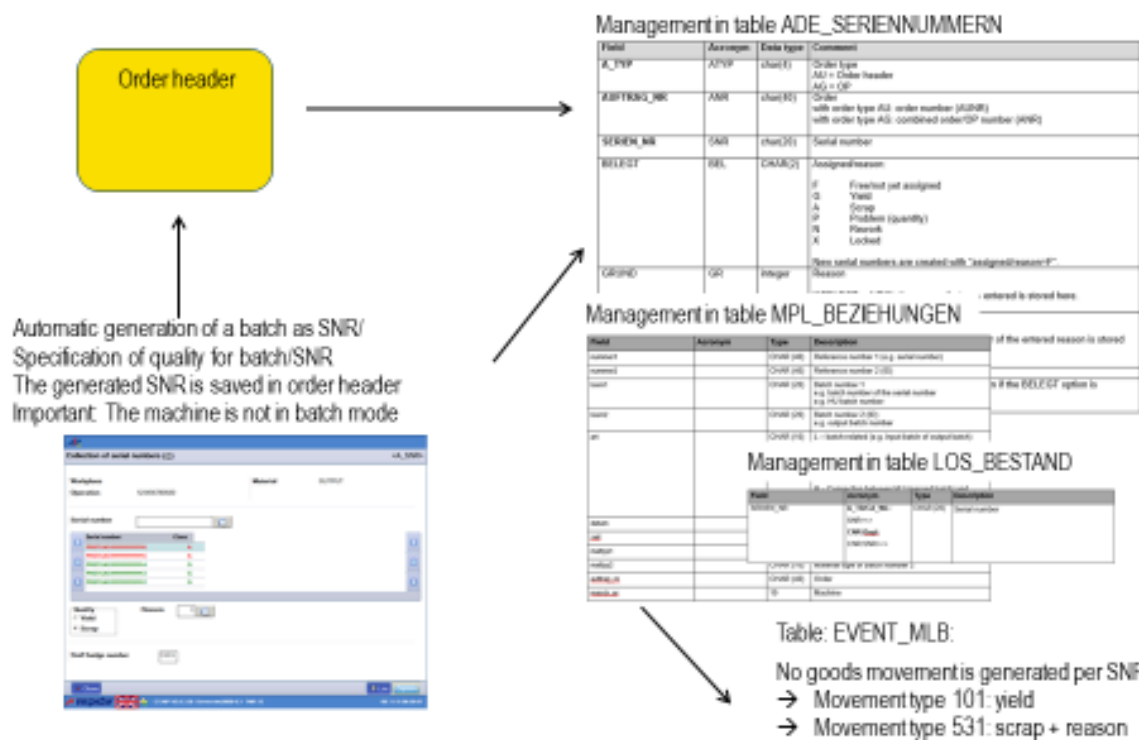
Exactly one batch with quantity 1 is created for each serial number. In this case, the serial number and the batch number are identical.

The "list" function updates the list of already recorded serial numbers.

For OPs handled in batches, the batch attributes that might have to be recorded are directly entered in the terminal dialog before confirming the terminal dialog.

A goods movement is generated for each serial number (please note: the machine mustn't be in batch mode!).

OP requiring batch mgmt. – A_SNR "G – automatic generation"



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mpcv

Collection of serial numbers (A SNR) including automatic assignment of serial numbers using the number range - type "S"

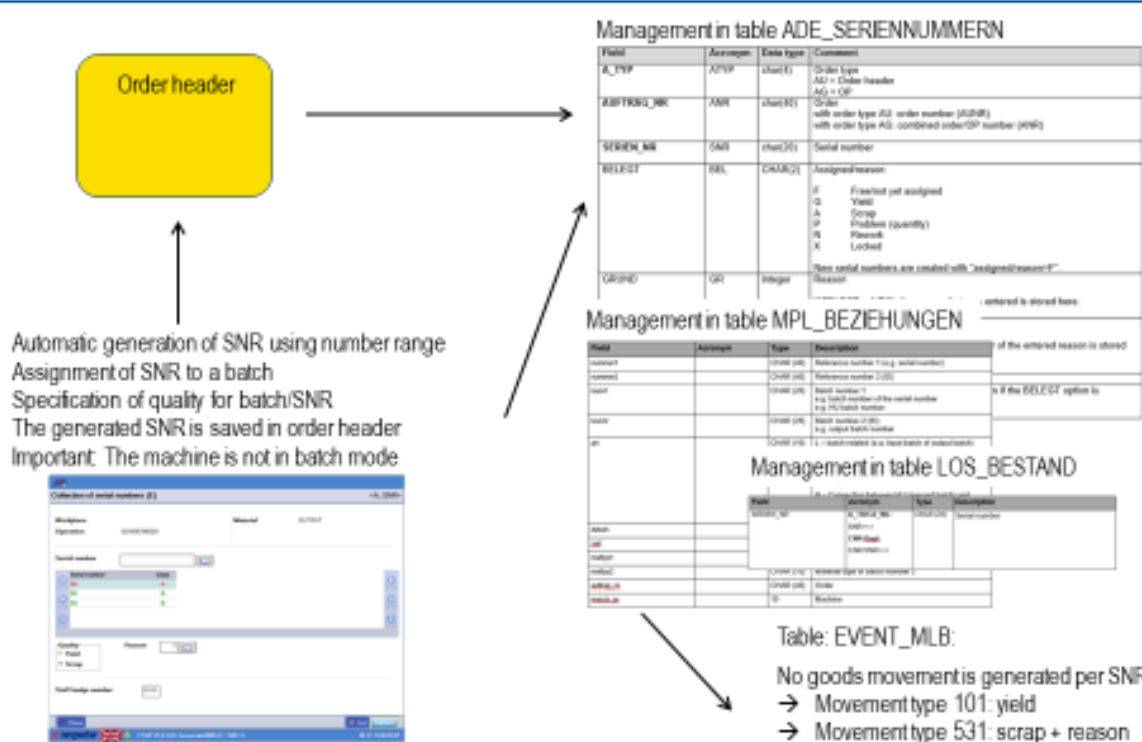
Serial numbers are recorded in relation to the batch if the option "serial number requirement = S" is set for the operation. In this case, the operation must be subject to batch management. Every registered serial number causes an output batch to be changed (CA_WL). This output batch change creates a batch with quantity 1 in the MPL module. This batch is connected with the serial number.

The server determines a new serial number and displays this number in the "serial number" field by clicking the function . The new serial number is assigned uniquely throughout the system via the "SNR" number range.

For OPs handled in batches, the batch attributes that might have to be recorded are entered in the general standard dialog before confirming/sending the data.

A goods movement is generated for each serial number (please note: the machine mustn't be in batch mode!).

OP requiring batch mgmt. – A_SNR "S – automatic generation/number range"



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Tables used to manage serial numbers

Table: AUFTRAGSBESTAND to configure the operation for serial number recording.

Field	Acronym	Type	Description
SERIENNR_PFLICHT	OPT:SNR=<>	CHAR (1)	Also see order type and/or processing code " / "N" = No serial number requirement "+" = Positive → "-" = Negative → "G" = Automatic generation of the serial number (only in combination with MPL as batch number) "S" = Automatic generation of the serial number using number range (only in combination with MPL as batch number) "E" = Manual collection of the serial number

Table: ADE_SERIENNUMMERN to manage serial numbers.

Field	Acronym	Data type	Comment
A_TYP	ATYP	char(4)	Order type AU = Order header AG = OP
AUFTRAG_NR	ANR	char(40)	Order with order type AU: order number (AUNR) with order type AG: combined order/OP number (ANR)
SERIEN_NR	SNR	char(20)	Serial number
BELEGT	BEL	CHAR(2)	Assigned/reason: F Free/not yet assigned G Yield A Scrap P Problem (quantity) N Rework X Locked New serial numbers are created with "assigned/reason=F".
GRUND	GR	integer	Reason If BELEGT = A/P/N, the reason that was entered is stored here.
GRUNDTXT_NR	GRTXTNR	integer	Reason text number If BELEGT = A/P/N the reason text number of the entered reason is stored here.
VERWEIS	VERWEIS	Series	Internal reference number
BEARB		CHAR(10)	The last modification is recorded here, even if the BELEGT option is changed.
BEARB_DAT		DATE	
BEARB_ZEIT		TIME	

Table: LOS_BESTAND (only for OPs that are subject to batch management) to manage the created batches.

Field	Acronym	Type	Description
SERIEN_NR	A_TR/CA_WL: SNR=<> CNR-Bapi: CNR:SNR=<>	CHAR (20)	Serial number

Table: MPL_BEZIEHUNGEN (only for OPs that are subject to batch management) to manage batches/
HU batches

Field	Acronym	Type	Description
nummer1		CHAR (40)	Reference number 1 (e.g. serial number)
nummer2		CHAR (40)	Reference number 2 (ID)
losnr1		CHAR (20)	Batch number 1 e.g. batch number of the serial number e.g. HU batch number
losnr2		CHAR (20)	Batch number 2 (ID) e.g. output batch number
art		CHAR (10)	L – batch-related (e.g. input batch of output batch) S – serial number (e.g. output batch of included serial number) B – connection on the same level (e.g. parallel output batches) H – Connection between HU (merged batch) and individual batch (e.g. HU and output batch)
datum		DATE	Entry date
zeit		TIME	Entry time
mattyp1		CHAR (10)	Material type of batch number 1
mattyp2		CHAR (10)	Material type of batch number 2
auftrag_nr		CHAR (40)	Order
masch_nr		10	Machine

ERP interface

Download of serial numbers

By default, serial numbers are not downloaded with reference to the order header. A customization is required in this case.

Upload of serial numbers:

For each recorded serial number, the quantity is uploaded to the ERP (interface EIS-ERP) in relation to the operation.

Additionally, a goods movement is generated for the recorded serial numbers. This goods movement can be transferred as goods receipt to the ERP via the [material interface](#) (EIS-MCL).

Application-relevant HYDRA settings

The following document describes the application-relevant settings and configurations in HYDRA for the collection and evaluation of serial numbers:

[Configuration of the collection and evaluation of serial numbers](#)